This article was downloaded by: [137.189.217.69] On: 05 October 2025, At: 06:53
Publisher: Institute for Operations Research and the Management Sciences (INFORMS)
INFORMS is located in Maryland, USA

Manufacturing & Service Operations Management

Publication details, including instructions for authors and subscription information:
http://pubsonline.informs.org

Improved Crop Productivity Through Optimized Planting Schedules

Ying Zhang, Jayashankar M. Swaminathan

To cite this article:
Ying Zhang, Jayashankar M. Swaminathan (2020) Improved Crop Productivity Through Optimized Planting Schedules. Manufacturing & Service Operations Management 22(6):1165-1180. https://doi.org/10.1287/msom.2020.0941

Full terms and conditions of use: https://pubsonline.informs.org/Publications/Librarians-Portal/PubsOnLine-Terms-and-Conditions

This article may be used only for the purposes of research, teaching, and/or private study. Commercial use or systematic downloading (by robots or other automatic processes) is prohibited without explicit Publisher approval, unless otherwise noted. For more information, contact permissions@informs.org.

The Publisher does not warrant or guarantee the article's accuracy, completeness, merchantability, fitness for a particular purpose, or non-infringement. Descriptions of, or references to, products or publications, or inclusion of an advertisement in this article, neither constitutes nor implies a guarantee, endorsement, or support of claims made of that product, publication, or service.

Copyright © 2020, INFORMS

Please scroll down for article—it is on subsequent pages

With 12,500 members from nearly 90 countries, INFORMS is the largest international association of operations research (O.R.) and analytics professionals and students. INFORMS provides unique networking and learning opportunities for individual professionals, and organizations of all types and sizes, to better understand and use O.R. and analytics tools and methods to transform strategic visions and achieve better outcomes.
For more information on INFORMS, its publications, membership, or meetings visit http://www.informs.org

http://pubsonline.informs.org/journal/msom

MANUFACTURING & SERVICE OPERATIONS MANAGEMENT
Vol. 22, No. 6, November–December 2020, pp. 1165–1180
ISSN 1523-4614 (print), ISSN 1526-5498 (online)

Downloaded from informs.org by [137.189.217.69] on 05 October 2025, at 06:53 . For personal use only, all rights reserved.

Improved Crop Productivity Through Optimized Planting Schedules

Ying Zhang,[a] Jayashankar M. Swaminathan[b]

[a] Department of Management, College of Business, Clemson University, Clemson, South Carolina 29634; [b] Kenan-Flagler Business School, University of North Carolina at Chapel Hill, Chapel Hill, North Carolina 27599

Contact: ying6@clemson.edu, https://orcid.org/0000-0002-4335-3602 (YZ); msj@unc.edu, https://orcid.org/0000-0002-1224-5422 (JMS)

Received: September 8, 2019
Revised: June 25, 2020; August 27, 2020
Accepted: August 28, 2020
Published Online in Articles in Advance: October 19, 2020

https://doi.org/10.1287/msom.2020.0941

Copyright: © 2020 INFORMS

Abstract. ***Problem definition***: We study the optimal seeding policy under rainfall uncertainty in rain-fed agriculture and explore its advantage over commonly used heuristics in practice. ***Academic/practical relevance***: Our work is in the area of agriculture operations, and we focus on the improvement of farmer's expected total profit by optimizing planting schedules. ***Methodology***: We model a farmer's planting problem under limited planting capacity in a finite-horizon stochastic dynamic program. ***Results***: We show that the optimal planting policy is a time-dependent, threshold-type policy, and the optimal threshold is dependent on the soil water content and planting capacity. In our computational study, we use the well-known Decision Support System for Agrotechnology Transfer simulator used in agriculture to obtain the expected yield when planting in any given period. Utilizing field weather data from Southern Africa, in a real-size, large-scale problem, we demonstrate a significant relative profit advantage of the optimal planting schedule over commonly used heuristics in practice. We show that the relative advantage of the optimal policy increases as the climate condition becomes more severe for planting. We also develop a heuristic based on the secretary problem and demonstrate the increased efficacy of the secretary heuristic. ***Managerial implications***: We show that the farmer should plant down to the optimal threshold of seed amount. However, in practice, farmers start to plant each year after observing enough cumulative rainfall. Utilizing field weather data, in a real-size, large-scale problem, we show significant improvement of the expected total profit if the farmer could adopt the optimal policy.

History: This paper has been accepted for the *Manufacturing & Service Operations Management* Special Issue on Responsible Operations and Supply Chain Management.
Supplemental Material: The online appendix is available at https://doi.org/10.1287/msom.2020.0941.

Keywords: optimal planting schedules • rainfall uncertainty • climate dynamics • crop yield simulation

1. Introduction

The demand for agricultural products is increasing because of world's growing population. By 2050, food production must double to meet the demand of the world (United Nations 2009). Increasing agricultural output is seen as a way out of poverty for the bottom of the pyramid. Governments and world organizations are working to create effective farm subsidies and minimum support prices (Tang et al. 2018). With roughly 60% of the global total uncultivated arable land, African countries are regarded as a critical player to feed the world. However, 80% of Africa's agriculture depends on rainfall as irrigation is too expensive for resource-poor farmers in this semiarid area (Obasanjo 2012). Because of outdated farming techniques and uncertainty in rainfall, crop yields in Africa are far below the developing world average. Africa generates only 10% of the global agricultural output (Lopes 2014), and as a result, hunger affects about 240 million African people (Obasanjo 2012). Furthermore, because of increased dry spells coupled with rain-fed irrigation, the ability to plant within the optimal time window has become extremely important (Kings 2019). Scientific agriculture in the form of effective planting, fertilizing, irrigation, pest control, and mechanization could potentially help farmers achieve higher yields.

Farm labor deficiency results in reduced agricultural productivity and severe food insecurity for farmers in sub-Saharan Africa (Food and Agriculture Organization 2009). Small-scale farmers rely heavily on manual labor for farming, lacking mechanization. Because small farmers also seek off-farm employment due to food insecurity from their farmland, they are unable to plant crops within the best planting window, and this leads to delayed planting (Clark et al. 2009). Medium and larger farmers often are not able to hire enough external labor to achieve timely sowing as the labor demand peaks during the busy planting period but the available labor does not.

Downloaded from informs.org by [137.189.217.69] on 05 October 2025, at 06:53 . For personal use only, all rights reserved.

Missing the best planting time because of the farm labor shortage during planting significantly harms the farm output. Therefore, increasing farming capacity, such as the introduction of planting mechanization, is critical and in urgent need to boost agricultural productivity (Food and Agriculture Organization 2009).

Many multinational firms in the agriculture industry are striving to develop innovative solutions to help small- and medium-scale farmers become more effective. Our motivation for this work comes from interactions with AGCO, the third largest manufacturer of farm equipment in the world. AGCO is introducing modern farming practices in Africa. In its efforts to introduce mechanization, it needs to quantify the benefits of scientific farming to funding agencies that could then finance those efforts. In order to do that, one needs to understand the optimal approach to planting seeds under uncertain rainfall and compare the benefits in relation to commonly used heuristics in practice.

Although labor capacity constraints are also felt in other agricultural operations, including fertilizing, irrigation, weeding, and harvesting, the planting schedule directly impacts the timing of those other constraints because planting schedules dictate other farming operations. Therefore, in this paper, in order to explore the impact of planting mechanization and optimal planting policy on the improvement of farmer's expected total profit, we model such labor constraints through planting capacity and demonstrate the relative advantage of mechanization and optimal policy under varying capacities in our computational study.

In this paper, we model a farmer's planting problem for a single crop under rainfall uncertainty as a finite-horizon stochastic dynamic program. In our model, a farmer could plant up to m batches of seeds in each period, where m is the planting capacity, and the farmer needs to decide how many batches of seeds to plant in each period given the soil water content in order to maximize the expected total profit from crop production. We show that the structure of the optimal schedule is a time-dependent, threshold-type policy in which the farmer should plant down to the optimal threshold; the remaining amount of seeds after planting should reach the optimal threshold unless limited by the planting capacity. This optimal plant-down-to threshold depends on the soil water content, planting capacity, and remaining time in the horizon, and it is nondecreasing in planting capacity. Furthermore, we provide conditions under which the threshold is nonincreasing in soil water content.

In our computational study, we utilize field weather data from Southern Africa to investigate the relative advantage of the optimal planting schedule over commonly used heuristics in practice and the impact of climate conditions on this advantage. The expected yield when planting in any given period is obtained using the well-known Decision Support System for Agrotechnology Transfer (DSSAT) simulator used in agriculture (Jones et al. 2003). The relative advantage of the optimal schedule varies with the initial soil water content and could be as high as 8.27%. Even when the initial soil water content is very low, the relative advantage of the optimal schedule is 6.68%. We further compare planting patterns under the optimal policy and commonly used myopic heuristics. When both mechanization and an optimal planting schedule are adopted, we find that the expected profit can be improved significantly by about 200% over existing planting heuristics and manual operations. We also develop a heuristic based on the traditional secretary problem and demonstrate that it outperforms the commonly used heuristics in almost all cases. Generally, farmers' profits suffer significantly from crop yield loss when the main rainfall starts later (Stern and Cooper 2011), when the expected length of dry spell before the main rainfall becomes longer (Dennett 1987), or when the within-season variability of rainfall becomes higher (Stern and Cooper 2011). We find that the advantages of the optimal planting schedule are higher under these conditions. This indicates that the adoption of the optimal planting schedule could mitigate farmers' risk of reduced profit resulting from severe climate conditions.

The rest of the paper is organized as follows. Section 2 discusses the related literature. Section 3 studies the crop planting model. In Section 4, we conduct an extensive computational study and explore the relative advantage of the optimal planting schedule over commonly used heuristics in practice. In Section 5, we present model variants that consider the fixed cost of planting and rainfall occurrence modeled as a discrete time Markov chain (DTMC). We conclude in Section 6.

2. Related Literature

Our work is in the area of agricultural operations. Lowe and Preckel (2004) review applications of planting models and decision technology to agriculture problems related to operations management. Recent papers in agricultural operations study irrigation resource allocation (Dawande et al. 2013), harvest risk (Allen and Schuster 2004), capacity and production planning with random yield and demand (Kazaz 2004, Kazaz and Webster 2011, Tan and Çömden 2012, Hu et al. 2019), crop planning (Maatman et al. 2002, Boyabatlı et al. 2019), food gleaning operations (Ata et al. 2019), government policy (Alizamir et al. 2019), and the agriculture market in developing economies (Tang et al. 2015). Among these, Tan and Çömden (2012) study the optimal farm area and seeding time of multiple farms to maximize the profit under uncertain demand.

Downloaded from informs.org by [137.189.217.69] on 05 October 2025, at 06:53 . For personal use only, all rights reserved.

The unit crop yield is modeled to be purely dependent on the seeding time for a specific farm. Kazaz (2004) and Kazaz and Webster (2011) study pricing and production planning in a two-stage stochastic programming framework in which the crop yield is dependent on the seeded amount. These papers do not study planting schedules and ignore the impact of uncertain rainfall on the crop profit. In contrast, we focus on the seed planting process under uncertain rainfall in rain-fed agriculture and model the seeding problem in a finite-horizon stochastic dynamic program. Maatman et al. (2002) model farmers' strategies of production, consumption, selling, purchasing, and storage in a two-stage stochastic programming framework with the objective to minimize deficits of various nutrients over multiple farming seasons. The production decision is dependent on observed rainfall that determines the number of days available for sowing. In our model, the objective is to maximize the farmer's expected total profit, and we establish the optimal seeding policy. The seeding decision is dependent on the uncertain rainfall as the rainfall determines the forecast of the crop yield. Ata et al. (2019) study the dynamic staffing policy in gleaning operations under uncertain food and labor supply and show that the optimal policy is a nested threshold policy. In the context of farming operations, we study the optimal seeding policy under uncertain rainfall (water supply) with the objective to maximize the expected total crop profit and show that the optimal policy is a threshold-type policy.

Production scheduling in the manufacturing industry has been extensively studied in operations management (Graves 1981). Most of the production scheduling models focus on minimizing total inventory cost during the planning horizon. In each period, inventory cost is incurred because of leftover inventory or unsatisfied demand that carries over to the next period. The optimal schedule minimizes the total expected cost. For the planting schedule problem, however, seeds can only be planted in sunny periods, and in each period, newly planted seeds generate the expected profit based on expected crop production, which depends on the current soil water content and rainfall, temperature, and solar radiation in the remaining periods of the planting horizon. The optimal planting schedule maximizes the expected total profit at the end of the planting horizon.

In agriculture literature, searching for the optimal planting dates for a single crop in rain-fed agriculture in Africa has been studied (Wolf et al. 2015, Srivastava et al. 2016). Using crop yield simulators, these papers focus on the search for the onset of the rainy season and, hence, the planting date. However, these papers ignore the stochasticity of the rainfall as well as planting capacity constraints faced by farmers. We use the finite-horizon stochastic dynamic program framework to model the farmer's planting decisions. Our model considers uncertain rainfall occurrence and stochastic precipitation amount and assumes that farmers only plant during sunny days. Therefore, to obtain the highest expected profit with a certain amount of seeds, our model searches for the optimal schedule of planting operations rather than a one-shot planting decision. Our paper is closely related to Swaminathan and Zhang (2020), who study a farmer's seeding schedule under rainfall uncertainty by modeling cumulative biomass production, utilizing earlier work in plant physiology. Instead of using a biomass model, we use the DSSAT model to compute the expected yield and also develop a modified version of the secretary heuristic.

3. Crop Planting Model

In this section, we present a farmer's planting model for a single crop in a finite horizon. A planting horizon consists of N periods with reverse time indexing, that is, the first period is period N, followed by $N-1$, $N-2$, and so on. The weather in the planting horizon is characterized by vector $\mathbf{p}^r$. The probability that it is rainy in period t is p_t^r. The probability that it is sunny in period t is $1-p_t^r$. As African countries receive many hours of sunshine on average and high-intensity solar radiation, we assume that the weather is sunny when it does not rain (SOLA 2013). We assume that, at the beginning of any period t, the farmer knows whether period t will be rainy or not. This assumption is reasonable because local weather forecast information is available to most farmers nowadays. The farmer cannot plant in rainy periods, and thus, the farmer needs to make the planting decision in each sunny period. Note that, for simplicity, we assume that each period represents one day. However, our model applies to general cases in which one period represents half a day or smaller intervals as well. We assume a fixed amount of seeds at the beginning of the planting horizon as most smallholder farmers in Africa use farm-saved seeds and that the farmer would not purchase additional seeds during the planting horizon (McGuire 2016).

We assume that fertilizing, pest control, weeding, and harvesting processes are automatically optimized by the farmer given the planting schedule and do not explicitly model these decisions. Fertilizers are often used to strengthen the root and leaf growth, blossom formation, and fruit production. Therefore, the effectiveness of fertilizing is highly dependent on the planting time and growth stage of crops (Grant 2018). Similarly, the schedule of insecticide use is dependent on the planting date because treatments are required to target specific growth stages and a time window

Downloaded from informs.org by [137.189.217.69] on 05 October 2025, at 06:53 . For personal use only, all rights reserved.

shortly before or after the planting date (Allen et al. 2017). Weeding is conducted in the early stage of crop growth to avoid competition with crops, and thus, the weeding schedule is dictated by the planting schedule. Harvesting is often scheduled to start sometime after the planting date, and the time gap between planting and harvesting is determined by the crop species and geographical characteristics (National Agricultural Statistics Service 2010). Therefore, scheduling any of these operations is dependent on and needs to be coordinated with the planting schedule.

Let m denote the planting capacity, $m \in \mathcal{N}^+$, and $s_t^m = (sd_t, sw_t)$ denote the system state under planting capacity m at the beginning of period t, $s_t^m \in \{(sd, sw) | sd \in \mathcal{N}, sw \in \mathcal{R}^+\}$. sd_t denotes the number of seed batches available for future planting in period t, and sw_t denotes the soil water content at the beginning of period t. Without loss of generality, we measure the seed amount by batches, and the batch size is dependent on the actual planting practice. Let i_t denote the farmer's planting decision in period t given it is sunny. Under planting capacity m, the farmer can plant $i_t = 0, 1, \ldots, \min\{m, sd_t\}$ batches of seeds in a sunny period t. We use $Q_t(sw_t)$ to denote the unit expected crop production of planting in period t: if one batch of seeds is planted in a sunny period t at soil water content sw_t, the expected crop production at harvesting is $Q_t(sw_t)$. In our computational study, we compute $Q_t(sw_t)$ using DSSAT: a crop growth simulation software that has been widely used for many crops under various environmental conditions by other researchers in agriculture and world organizations such as the Food and Agriculture Organization (Jones et al. 2003) (see Section 4.2). As we focus on the improvement of planting operation timeliness by the optimal planting schedule and mechanization, we assume that all seeds that are planted in the same period are identical in expected crop production and ignore the difference in the unit expected crop production of planting ($Q_t(sw_t)$) when the planting capacity varies. If i_t batches of seeds are planted in period t, the expected crop production of these seeds at harvesting is $i_t \times Q_t(sw_t)$. We use $\omega_t(.)$ to denote the transition function of soil water content, $prec_t$ to denote the realized value of stochastic conditional precipitation in period t given it rains, and $\boldsymbol{\lambda}_t$ to denote the vector of other environmental and soil-related parameters that determine the dynamics of soil water content in period t. As the change of the soil water content is largely driven by the precipitation for a given type of soil and geographical location, we consider $\omega_t(.)$ as a function of the soil water content (sw_t), realized total precipitation ($prec_t$), and other parameters ($\boldsymbol{\lambda}_t$) in period t. Given that the soil water content at the beginning of period t is sw_t, the soil water content at the beginning of period $t-1$ is $sw_{t-1} = \omega_t(sw_t, 0, \boldsymbol{\lambda}_t)$ if it does not rain in period t and $sw_{t-1} = \omega_t(sw_t, prec_t, \boldsymbol{\lambda}_t)$ if it rains in period t and the realized total precipitation in period t is $prec_t$. As the soil water content is dependent on various parameters and modeling the soil water content is complicated, we do not specify the form of $\omega_t(.)$ in our model and use DSSAT to generate sample path values of the soil water content in our computational study when other variables, such as soil features, temperature, and solar radiation are factored in (see Section 4.2).

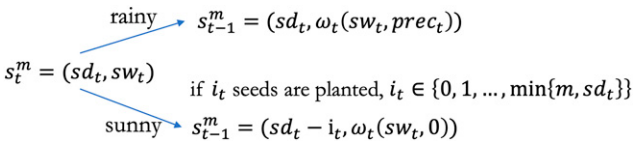
Table 1 summarizes our notations, and Figure 1 illustrates the state transition. At the beginning of period t, if it is rainy, the farmer cannot plant, and no decision needs to be made. The number of seed batches available for future planting remains the same until the beginning of period $t-1$. If it is sunny in period t, the farmer needs to decide how many batches of seeds to plant. If i_t batches of seeds are planted in period t, the number of seed batches available on hand would decrease by i_t, $sd_{t-1} = sd_t - i_t$. Let $G_t(sw_t)$ denote the farmer's expected profit from planting one batch of seeds in period t at soil water content sw_t. Suppose the farmer's income is p per unit weight of crop production and the variable cost of crop cultivation is c per batch of seeds. Then, $G_t(sw_t) = p \times Q_t(sw_t) - c$. If i_t batches of seeds are planted in period t, they contribute expected profit of $i_t \times G_t(sw_t)$ at the end of the planting horizon. Our objective is to find a planting schedule that maximizes the farmer's expected total profit at the end of the planting horizon. We use r_v to denote the value of one batch of seeds left unplanted at the end of the planting horizon and $V_t^m(.)$ to denote the farmer's maximum

Table 1. Notations

Notation	Explanation
m	Planting capacity: maximum number of seed batches the farmer can plant in a sunny period
sd_t	Number of seed batches available for future planting at the beginning of period t
p_t^r	Probability of rainfall occurrence in period t
sw_t	Soil water content at the beginning of period t
$prec_t$	Realized value of stochastic conditional precipitation amount in period t given it rains
$\boldsymbol{\lambda}_t$	The vector of environmental and soil-related parameters other than sw_t and $prec_t$ that determine the dynamics of soil water content in period t
$\omega_t(.)$	Transition of soil water content from period t to $t-1$ as a function of soil water content (sw_t), realized total precipitation amount ($prec_t$, zero if sunny) and other parameters ($\boldsymbol{\lambda}_t$) in period t
i_t	Number of seed batches that the farmer decides to plant in period t given it is sunny
$Q_t(.)$	Expected crop production by planting one batch of seeds in period t
$G_t(.)$	Expected profit by planting one batch of seeds in period t
$V_t^m(.)$	Maximum expected total profit with t periods to go under planting capacity m

Downloaded from informs.org by [137.189.217.69] on 05 October 2025, at 06:53 . For personal use only, all rights reserved.

Figure 1. (Color online) State Transition for Crop Planting Model



expected total profit with t periods to go as a function of system state $s_t^m = (sd_t, sw_t)$. Then,

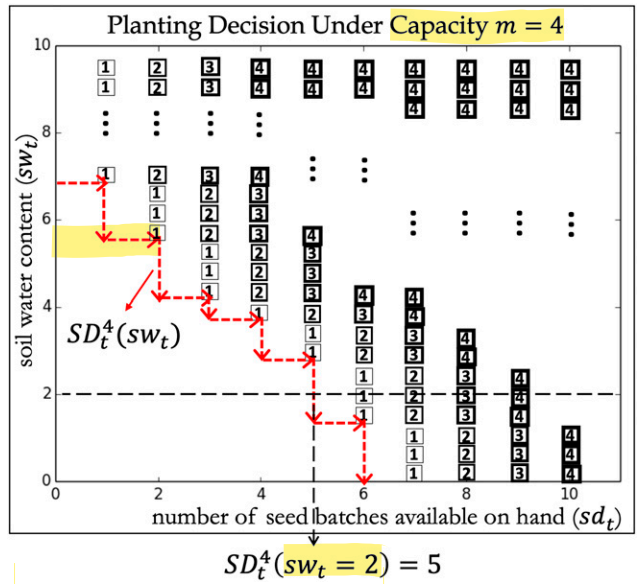
$$\begin{aligned} V_t^m(sd_t, sw_t) = {} & p_t^r \mathbb{E}_{prec_t} V_{t-1}^m(sd_t, \omega_t(sw_t, prec_t, \lambda_t)) \\ & + (1 - p_t^r) \max_{i_t = 0,1,\ldots,\min\{sd_t, m\}} \{V_{t-1}^m(sd_t - i_t, \\ & \omega_t(sw_t, 0, \lambda_t)) + i_t \times G_t(sw_t)\}, \ t = 1, \ldots, N \\ V_0^m(sd_0, sw_0) = {} & r_v \times sd_0. \end{aligned} \tag{1}$$

Theorem 1. *Let $a_t^m(sd_t, sw_t)$ denote the optimal planting decision for a sunny period t under capacity m. Given state $s_t^m = (sd_t, sw_t)$ in period t, there exists a threshold $SD_t^m(sw_t)$ that the optimal decision $a_t^m(sd_t, sw_t)$ satisfies*

$$a_t^m(sd_t, sw_t) = \begin{cases} \min\{m, sd_t - SD_t^m(sw_t)\} & \text{if } sd_t > SD_t^m(sw_t) \\ 0 & \text{otherwise.} \end{cases} \tag{2}$$

Starting to plant early ensures that more seeds can be planted throughout the planting horizon. However, the low soil water content at planting may result in low crop yields ($Q_t(sw_t)$). On the other hand, planting seeds late at a higher soil water content brings high crop yields although this results in fewer seeds being planted because of limited planting capacity. The optimal planting schedule is a time-dependent, threshold-type policy, and Figure 2 shows an example of the optimal planting policy in period t. Given the planting capacity $m = 4$ and the soil water content in period t, $sw_t = 2$, the optimal decision is not to plant when the farmer has no more than five batches of seeds ($a_t^4(sd_t, 2) = 0$ for $sd_t \le 5$). If the farmer has six to nine batches of seeds, the optimal decision is to plant one to four batches of seeds, respectively ($a_t^4(sd_t, 2) = sd_t - 5$ for $6 \le sd_t \le 9$). When the farmer has more than nine batches of seeds, the optimal decision is to plant four batches of seeds as the planting capacity is four and no more than four batches of seeds can be planted in a single period ($a_t^4(sd_t, 2) = 4$ for $sd_t > 9$). In this case ($sw_t = 2, m = 4$), the optimal decision in period t can be described by a threshold $SD_t^4(sw_t = 2) = 5$ that is dependent on the soil water content in period t and planting capacity. In sunny period t, if the number of seed batches is larger than the threshold $SD_t^4(sw_t = 2)$, the optimal decision

Figure 2. (Color online) An Example of Optimal Planting Policy Under Capacity $m = 4$ with t Periods to Go



is to plant down to $SD_t^4(sw_t = 2)$ unless limited by the capacity. Otherwise, the optimal decision is to hold seeds until the next period. As a result, as long as the planting capacity allows and the seed amount is above the planting threshold, if the number of seed batches available on hand increases by one, the optimal planting decision also increases by one batch. This result is similar to the modified base stock policy for inventory problems under capacity constraint (Federgruen and Zipkin 1986).

Proposition 1. *Assume $G_t(sw_t) > 0$ and $\omega_t(sw_t, prec_t, \lambda_t)$ is nondecreasing in sw_t, $t = 1, \ldots, N$. $SD_t^m(sw_t)$ is nonincreasing in sw_t if (1) $\frac{G_k(sw_k)}{G_{k+1}(sw_k)}$ is nonincreasing in sw_k for any $sw_k \ge 0$, $k = 1, \ldots, t-1$ and (2) $\frac{G_k(\omega_k(sw_k, prec_k, \lambda_k))}{G_k(sw_k)}$ is nonincreasing in sw_k for any $sw_k, prec_k \ge 0$ and λ_k, $k = 1, \ldots, t$.*

Proposition 1 provides conditions on the expected profit from planting one batch of seeds ($G_t(.)$) under which the optimal planting threshold $SD_t^m(sw_t)$ is nonincreasing in the soil water content, that is, the farmer is willing to plant more at a higher soil water content. Generally, under an arid and semiarid climate as in African countries, higher soil water content leads to higher expected profit of one batch of seeds, no matter if the seeds are planted in the current period ($G_t(sw_t)$) or reserved for the next period ($V_{t-1}^m(sd_t, \omega_t(sw_t, 0, \lambda_t)) - V_{t-1}^m(sd_t - 1, \omega_t(sw_t, 0, \lambda_t))$). The conditions in Proposition 1 guarantee that, at higher soil water content, the expected profit of planting one batch of seeds in the current period ($G_t(sw_t)$) increases relatively more than that of planting in the next period ($G_{t-1}(\omega_t(sw_t, 0, \lambda_t))$). Therefore, the farmer is willing to plant more at a higher soil water content.

Downloaded from informs.org by [137.189.217.69] on 05 October 2025, at 06:53 . For personal use only, all rights reserved.

Proposition 2. *$SD_t^m(sw_t)$ is nondecreasing in m. Furthermore, $SD_t^m(sw_t) + 1 \leq SD_t^{m+1}(sw_t)$ when $SD_t^m(sw_t) > 0$.*

The optimal plant-down-to level is nondecreasing in the planting capacity m as stated in Proposition 2. Obviously, under infinite capacity, all seeds are planted in a single day to maximize the expected total profit (details in Section 3.2). Under limited planting capacity, however, not all seeds can be planted on the best planting day, harming the expected total profit. In order to mitigate the loss resulting from planting capacity limitation in future periods, the optimal planting threshold tends to be smaller under a lower planting capacity than a higher capacity. This result is consistent with the literature of inventory management under limited production capacity in which the order-up-to level tends to be nonincreasing in the capacity (Federgruen and Zipkin 1986).

3.1. Planting with Manual Labor

The planting capacity for small-scale farmers in Southern Africa is low as farmers rely heavily on manual labor for planting. Without loss of generality, we assume that the farmer's planting capacity is one ($m = 1$) with manual labor. As a special case of the crop planting model, the farmer's decision is whether to plant in each sunny period, and the optimal policy is still a time-dependent, threshold-type policy with the optimal threshold denoted by $SD_t(sw_t)$. With soil water content sw_t in sunny period t, if the farmer has more than $SD_t(sw_t)$ batches of seeds, the farmer should plant. Otherwise, the farmer should reserve the seeds for the next period.

3.2. Planting Under Infinite Capacity

Under unlimited planting capacity ($m \to \infty$), the planting problem becomes a special case of the crop planting model, and all seeds are planted in a single period as stated in Proposition 3. For a given sunny period, the farmer needs to decide whether to plant all seeds in the current period or reserve all seeds for the next period.

Proposition 3. *Under infinite planting capacity, the optimal planting behavior would be to plant all seeds in a single period.*

The infinite capacity problem is similar to the traditional secretary problem in which the employer determines the optimal stopping time of interviews and accepts the current interviewee to fit the position (Freeman 1983). In the planting schedule problem under infinite capacity, planting occurs if the expected profit of planting one batch of seeds is higher than that when this batch of seeds is reserved for the next period. However, for the secretary problem, the optimal policy is to pass the first N/e interviewees (N is the total number of interviewees, and e is the natural logarithm) and accept the next interviewee whose value is higher than all previous interviewees. Two fundamental differences in the two problems lead to the difference in the optimal policy. First, in the planting problem, the expected profit when a batch of seeds is planted in the current period or reserved for the next period can be forecasted by prior weather information about the remaining days of the planting horizon, but in the secretary problem, the interviewer has no information about the value of future interviewees. In addition, in the planting problem, the number of sunny periods when the farmer can plant is stochastic.

4. Computational Study

Our computational study aims to (i) investigate the benefit of using an optimal planting policy in comparison with commonly used heuristics in practice, (ii) develop a new heuristic for the planting problem and explore its performance, and (iii) identify climate conditions in which using the optimal policies might be more beneficial. We focus on the planting schedule of maize in Southern Africa, which is the staple crop in the area.

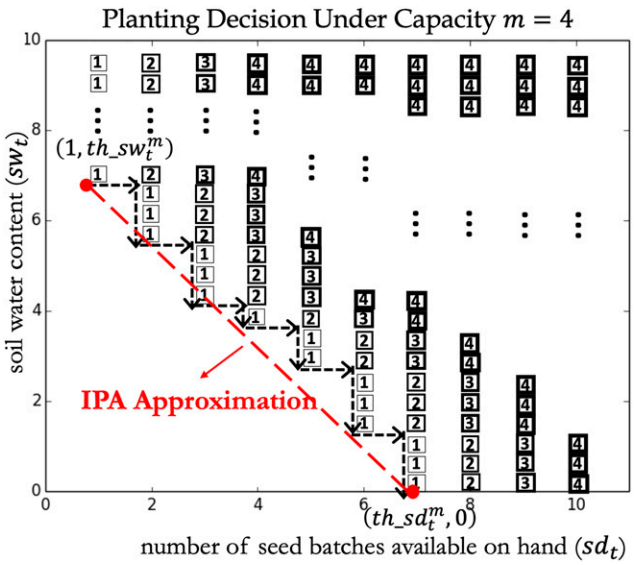
4.1. The Real-Size Problem

In the real problem that motivated this work, the rainy season typically goes from November to April in a six-month period. As crops need time to grow to maturity after planting, we consider the maize planting window to be from November to January (short season, 90-day cultivar). This is consistent with the practice in Southern Africa that crops need to be harvested before the frost comes in April (Kings 2019). The planting window spans from November 1 to January 31, thereby having 92 days in which planting decisions may need to be made.

Although we can use backward induction to search for the optimal planting schedule and obtain expected total profits, the running time increases exponentially in the length of the planting window. In order to handle such a large-scale problem and find the optimal solution, we develop an approximation based on infinitesimal perturbation analysis (IPA). According to Theorem 1, in each period, the optimal plant-down-to policy can be illustrated in a stair structure as represented by the arrows in Figure 3. For each period, we assume that $SD_t^m(sw_t)$ is a linear function of sw_t for a given value of m and use a line segment to approximate the stair structure as indicated by the dashed line in Figure 3. We then apply IPA to search for the optimal line segments associated with all periods under the assumption of continuous action space. Under planting capacity m, for period t, we use $th_sw_t^m$ to denote the highest level of soil water content

Downloaded from informs.org by [137.189.217.69] on 05 October 2025, at 06:53 . For personal use only, all rights reserved.

Figure 3. (Color online) Illustration of IPA Approximation



below which the farmer would not plant with one batch of seeds available when it is sunny and $th_sd_t^m$ to denote the largest number of seed batches available below which the farmer would not plant with the soil water content approaching zero when it is sunny. Therefore, the linear approximation of the threshold function can be defined by connecting two points $(1, th_sw_t^m)$ and $(th_sd_t^m, 0)$ as demonstrated in Figure 3. For period t, searching for the optimal line segment that approximates $SD_t^m(sw_t)$ is equivalent to searching for the optimal decision variables $(th_sw_t^m, th_sd_t^m)$ of the line segment, and we apply gradient search in this process. IPA guarantees that the expectation of sample-path gradient we obtain converges to the gradient of the expected total profit. Each time we evaluate the expected gradient of the total profit with respect to the threshold decision variables, we take the average of 500 sample-path gradients. Once the thresholds are searched out, the expected profit values are calculated by taking the average of 500 sample-path profit values, each obtained following the plant-down-to policy with thresholds computed.

To make sure that the convergence of IPA happens properly, we impose two stopping criteria in searching for the optimal thresholds. The first criterion is that, in two consecutive search iterations, if the absolute value of the relative average gradient change with respect to any decision variable is smaller than a threshold denoted by *grtol*, then we stop the search. The second criterion is that, in two consecutive search iterations, if the absolute value of the relative expected total profit change for any combination of initial seed amount and initial soil water content (sd_N, sw_N) is smaller than a threshold denoted by *tol*, then we stop the search. In our computational study, we use $grtol = 0.5\%$ and $tol = 0.5\%$. Setting this as the benchmark, we calculate the relative deviation from the expected total profit values when $grtol \in \{1\%, 0.75\%, 0.25\%\}$ and $tol \in \{1\%, 0.75\%, 0.25\%\}$. Our results show that the average total profit values deviate minimally when the values of *grtol* and *tol* deviate from the benchmark setting, and the running time of the algorithm increases significantly when *grtol* or *tol* is reduced to 0.25%.

4.1.1. Commonly Used Myopic Heuristics. In Africa, farmers commonly start to plant after observing several consecutive days of rain or enough cumulative precipitation. For the staple product, maize, one of the rules of thumb is to start planting after observing 25 mm of cumulative precipitation in a 10-day period starting from November 1 and then keep planting at full capacity until all seeds are depleted or the end of the planting window, whichever occurs earlier (Tadross et al. 2009). The expected profit values under commonly used heuristics are obtained by taking the average of 500 sample-path profits, each obtained under the $10\,days - 25$mm heuristic. Note that under commonly used heuristics with planting capacity $m > 1$, when the decision is to plant, the amount of seeds planted is set to be the planting capacity m.

4.1.2. A Heuristic Based on the Secretary Problem. Under unlimited planting capacity ($m \to \infty$), the planting behavior is a one-shot decision that is similar to the traditional secretary problem (Freeman 1983). We develop a heuristic based on the optimal policy of the secretary problem and explore the performance of the secretary heuristic compared with the commonly used heuristics in Section 4.6. We assume that the farmer would wait for $(\frac{N^s}{e} - \frac{N^s}{4m})$ sunny days, where N^s is the expected number of sunny days in the planting window and e is the natural logarithm, and then start planting at full capacity (m) from the next sunny day until the end of the planting window or the depletion of seeds, whichever occurs earlier. The expected total profit values under the secretary heuristic are obtained by taking the average of 500 sample-path profits. With the planting capacity going to infinity ($m \to \infty$), the waiting time before planting start converges to the similar form of the optimal waiting time in the secretary problem $(\frac{N^s}{e} - \frac{N^s}{4m} \to \frac{N^s}{e})$.

4.2. Weather Data

We use daily weather data of 31 geographical locations that span nine countries in Southern Africa (Zambia, Malawi, Zimbabwe, Botswana, Mozambique, Namibia, South Africa, Lesotho, and Swaziland) from https://power.larc.nasa.gov. These locations are reported to be major provinces and areas of maize production in specific countries. As rain stops farmers from planting, we define a rainy day if the daily cumulative precipitation is higher than 2 mm. This is

Downloaded from informs.org by [137.189.217.69] on 05 October 2025, at 06:53 . For personal use only, all rights reserved.

because the average daily cumulative precipitation amount through a dry month is around 2 mm. We use the aggregated daily weather data (solar radiation, minimum and maximum temperature, and probability of rain for a specific date) in the rainy season, which is obtained by taking the average over all geographical locations and years. We use a gamma distribution ($\Gamma_t(\alpha_t, \beta_t)$) to fit the conditional precipitation amount for a specific date (month, day) given that it is rainy as the gamma distribution is the commonly used distribution to fit nonzero rainfall precipitation (Srikanthan and McMahon 2001). The gamma distribution is also used by the weather module of DSSAT (Jones et al. 2003).

4.2.1. Sample-Path Crop Production and Soil Water Content. For a sunny period t, the farmer's expected profit from planting one batch of seeds ($G_t(sw_t)$) is linear in the expected crop production of these seeds ($Q_t(sw_t)$). We use DSSAT to generate the sample-path crop production values ($Q_t(sw_t)$) and the evolution of soil water content (sw_t) used in our heuristics. Other than the weather data, input data such as soil type, planting spacing, and crop type (cultivar) are the same as reported by DSSAT applications in the Southern Africa area (Zinyengere et al. 2015). For each of the scenarios we study, 1,000 sample paths of rainfall (rainfall occurrence and precipitation amount) are generated based on the aggregated weather data (rainfall probability and distribution of conditional precipitation). For a specific day, other input weather data such as minimum and maximum temperature and solar radiation, remain the same throughout all sample paths. For each day t on a specific sample path, we obtain the predicted soil water content (sw_t) and the sample-path yield of planting one batch of seeds in day t ($Q_t(sw_t)$) from DSSAT simulations. We use the water content (mm^3/mm^3) at the top level soil ($0 \sim 5$ mm) in our heuristics as the water content at this level of soil is relatively sensitive to the precipitation amount, and changes of soil water content at different levels are consistent. The yield estimate results from DSSAT also show that the top layer of soil ($0 \sim 5mm$) has the most part of the root system of the plant, which absorbs water from the soil to support plant growth.

4.3. Relative Advantage of Optimal Planting Schedule Using Approximation

We consider the cases in which the planting capacity $m \in \{1, 2, 3, 4\}$ and compare the average profits under the optimal planting schedule over commonly used myopic heuristics in practice using field weather data. We use the relative advantage of average total profit under the optimal planting schedule over that under commonly used heuristics,

$$\frac{\text{Average Profit of Optimal Planting Schedule} - \text{Average Profit of Commonly Used Heuristics}}{\text{Average Profit of Commonly Used Heuristics}},$$

to measure the benefit from adopting the optimal planting schedule. As the farmer's expected total profit is linear in the cumulative crop production, we consider $p = 1$ and $c = 0$. We also discuss the case in which $c > 0$ and compare the farmer's expected total profits under the two policies in a later context. For planting capacity m, the relative profit advantage of the optimal policy is calculated by taking the average of expected total profits when the initial seed amount is between $(31 \times m)$ batches and $(40 \times m)$ batches. This is consistent with the number of sunny days during which farmers can plant. Our results show that this range of seed amount provides the lower bound of the relative profit advantage of the optimal policy throughout all initial seed amount scenarios considered in our study ($(1 \sim 92) \times m$). We consider three cases of initial soil water content from the soil water data: average soil water content ($0.199mm^3/mm^3$ on November 1) based on the current rainfall data, low soil water content ($0.095mm^3/mm^3$ on November 1) when it is assumed that the rainfall in October is the same as the average rainfall throughout the dry season, and high soil water content ($0.298mm^3/mm^3$ on November 1) when it is assumed that the rainfall in October is the same as the average of the peak rainfall month, January (all simulation experiments start on October 1 at soil water content $0.294mm^3/mm^3$).

Table 2 demonstrates the relative profit advantage of the optimal schedule over commonly used myopic heuristics when the planting capacity under both planting policies is the same. When the initial soil water content is low, the farmer's expected total profit gets hurt. In spite of that, adopting the optimal

Table 2. Relative Expected Total Profit Advantage of the Optimal Policy (Average Standard Errors of Sample Path Relative Profit Advantage of the Optimal Policy in Parentheses)

Planting capacity	1	2	3	4
Low soil water content, %	6.90 (0.64)	6.68 (0.63)	6.87 (0.53)	7.15 (0.57)
Average soil water content, %	6.97 (0.56)	6.68 (0.52)	7.20 (0.58)	7.23 (0.54)
High soil water content, %	7.91 (0.60)	7.56 (0.61)	8.27 (0.68)	8.03 (0.63)

Downloaded from informs.org by [137.189.217.69] on 05 October 2025, at 06:53 . For personal use only, all rights reserved.

schedule leads to improvement in the expected total profit even under low initial soil water content. Our results show that, although the relative profit advantage of the optimal policy is higher with higher initial soil water content, it is not influenced a great deal by the initial soil water content. This is because the impact of initial soil water content quickly diminishes after the start of the rainy season, and the initial soil water content only influences the crop production of seeds that are planted early in the rainy season. The relative profit advantage of the optimal policy under manual planting ($m = 1$) is 6.97% with average initial soil water content, 6.90% with low initial soil water content, and 7.91% with high initial soil water content. Under mechanized planting ($m > 1$), the relative profit advantage of the optimal policy minimally changes as the planting capacity changes. Our results show that, for each planting capacity $m \in \{1, 2, 3, 4\}$, the sample path relative profit advantage of the optimal policy over the myopic policy is significantly larger than zero (significance level 1%). We present the average standard errors of sample path relative profit advantage of the optimal policy when the initial seed amount is between $(31 \sim 40) \times m$ in Table 2 (values in parentheses). Our results show that the average standard error is around 0.60% regardless of the initial amount of seeds available for planting and initial soil water content. Furthermore, using the income and cost data of dryland maize in five areas in South Africa (https://www.opot.co.za/index.php?page=home-page), we take the average value of income and cost data at $p/c = 1.3730$ and evaluate the relative profit advantage of the optimal policy over the commonly used heuristics. With manual labor ($m = 1$), adopting the optimal policy could increase the farmer's profit by 8.92% (8.39%, 9.88%) with average (low, high) initial soil water content. We present the detailed results in the online appendix.

Under infinite planting capacity, the planting behavior would be planting all seeds in a single period (see Proposition 3), which is the same as the case in which the planting capacity is one ($m = 1$) and the initial seed amount is one ($sd_N = 1$). Hence, the relative profit advantage of the optimal policy under infinite planting capacity is calculated under the setting with planting capacity of one and initial seed amount of one batch ($m = 1, sd_N = 1$). Our results show that, under infinite planting capacity, the expected total profit under the optimal policy is significantly higher than that under the myopic policy (significance level 1%). The relative advantage of the optimal policy is 95.44% with average initial soil water content, 87.43% with low initial soil water content, and 98.21% with high initial soil water content. The impact of the initial soil water content on the relative advantage of the optimal policy appears more significant under infinite planting capacity than that under limited planting capacity.

Table 3 illustrates the relative profit advantage of mechanized ($m > 1$) or manual ($m = 1$) planting under the optimal policy over manual planting ($m = 1$) under commonly used myopic heuristics. When moving from manual planting ($m = 1$) to mechanized planting ($m > 1$), we are increasing the number of seed batches available to the farmer proportional to the increased speed. By adopting mechanization ($m = 3$) under the optimal policy, the expected total profit can be improved by nearly 200%. This improvement by adopting both the optimal policy and mechanization is consistent with the observations made in the controlled pilot study in Zambia by AGCO (Swaminathan 2018).

Note that this comparison does not take into account the cost associated with mechanization and corresponding increase in the number of seeds. However, based on the magnitude of benefits through mechanization, it is clear that government policy or interventions by firms such as AGCO, to enable it would have a major impact. An optimal policy based on soil water content (and other environmental factors) should be coupled with such effort.

4.4. Comparison of Planting Patterns Under Optimal Policy and Myopic Policy

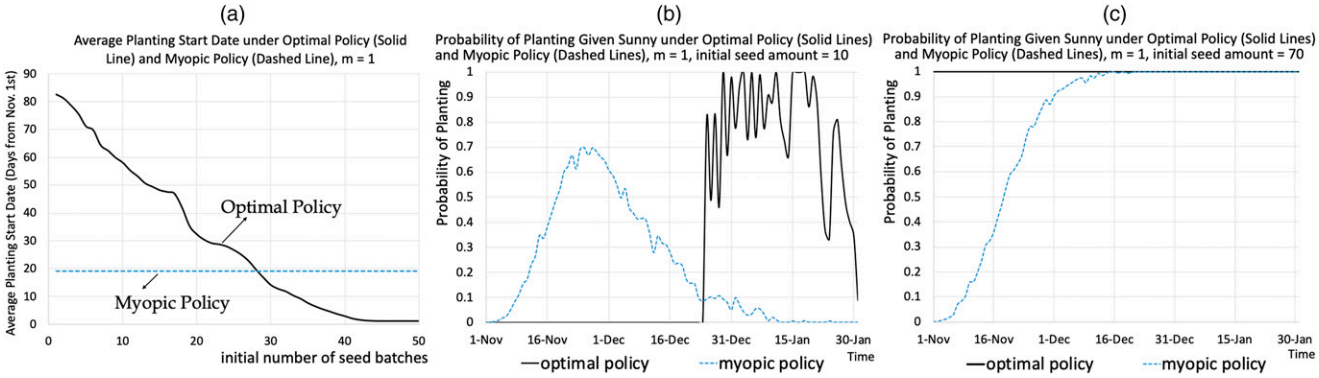
We further discuss the details of planting patterns under the optimal policy and myopic heuristics by exploring the average planting start date, probability of planting given sunny weather, and average amount of seeds left unplanted at the end of the planting window under the two policies. Figure 4(a) compares the average planting start dates (first day of planting) under the optimal policy (solid line) and myopic policy (dashed line) for manual planting ($m = 1$). The average planting start date is determined by the initial amount of seeds (sd_N) under the optimal policy but not under the myopic heuristics. Under the myopic heuristics, planting starts as soon as cumulative rainfall of 25 mm is observed in a 10-day window regardless of the amount of seeds on hand. Under the optimal policy, however, planting happens if the amount of available seeds is larger than the planting threshold. Hence, planting starts earlier with more seeds available to plant throughout the planting window.

Table 3. Relative Expected Total Profit Advantage of the Optimal Policy and Mechanization

Planting capacity	1	2	3	4
Low soil water content, %	6.90	97.22	197.25	295.32
Average soil water content, %	6.97	97.52	195.13	294.96
High soil water content, %	7.91	98.76	195.03	294.00

Downloaded from informs.org by [137.189.217.69] on 05 October 2025, at 06:53 . For personal use only, all rights reserved.

Figure 4. (Color online) Comparison of Planting Patterns Under the Optimal Policy and Myopic Heuristics



We compare the probability of planting given sunny weather under the optimal policy and the myopic heuristics when the initial seed amount is 10 batches (in Figure 4(b)) and 50 batches (in Figure 4(c)) for manual planting ($m = 1$). The x-axis represents days from the beginning of the planting window. The difference in the probability of planting between the optimal policy and myopic heuristics is dependent on the initial seed amount. When the initial seed amount is small ($sd_N = 10$), the probability patterns under the two policies are quite different. The probability of planting under the optimal policy is much lower than that under the myopic policy in the early stage of the planting window but much higher in the later stage of the planting window because the farmer plants much later under the optimal policy than the myopic policy. When the initial seed amount is large ($sd_N = 50$), the probability of planting under the optimal policy is much higher than that under the myopic policy in the early stage of the planting window as planting starts earlier under the optimal policy and the probabilities of planting under the two policies are very close in the later stage of the planting window.

Our results show that fewer seeds are left unplanted under the optimal policy than the myopic heuristics unless the initial seed amount is small and no seeds are left unplanted. With the initial seed amount of 50 batches under manual planting ($m = 1$) and the average number of sunny days of 50.47, on average, 1.78 (3.56%) batches of seeds would be left unplanted under the optimal policy and 17.94 (35.89%) batches would be left unplanted under the myopic policy. This is because planting starts earlier under the optimal policy when the initial seed amount is high (see Figure 4(a)), whereas the myopic policy waits too long and, hence, is unable to utilize all the seeds. We present the detailed results in the online appendix.

4.5. Sensitivity Analysis of Soil Data Accuracy

As small-scale farmers in sub-Saharan Africa may not be able to acquire accurate soil moisture data, we conduct sensitivity analysis on the soil moisture data accuracy and explore the loss of expected total profit and change of planting decisions resulting from observation errors in the soil water content. For each day t in the planting window, we consider the soil water content with observation errors to be uniformly distributed between $(1-\eta)\times sw_t$ and $(1+\eta)\times sw_t$, where sw_t is the accurate soil water content in period t and $\eta \in (0,1)$ represents the observation error. For each $\eta \in \{0.1, 0.3, 0.5\}$, we generate new soil water content data based on 500 sample paths of accurate soil water content data and evaluate the optimal policy when the farmer makes planting decisions based on the new soil data with observation errors and using the planting thresholds searched before. Table 4 lists the relative loss of expected total profit resulting from observation errors in soil moisture data. Our results show that the observation errors in soil water content do not result in great loss of expected total profit. Even when the observation error is large at $\eta = 0.5$, the average relative loss of the expected total profit is less than 1%.

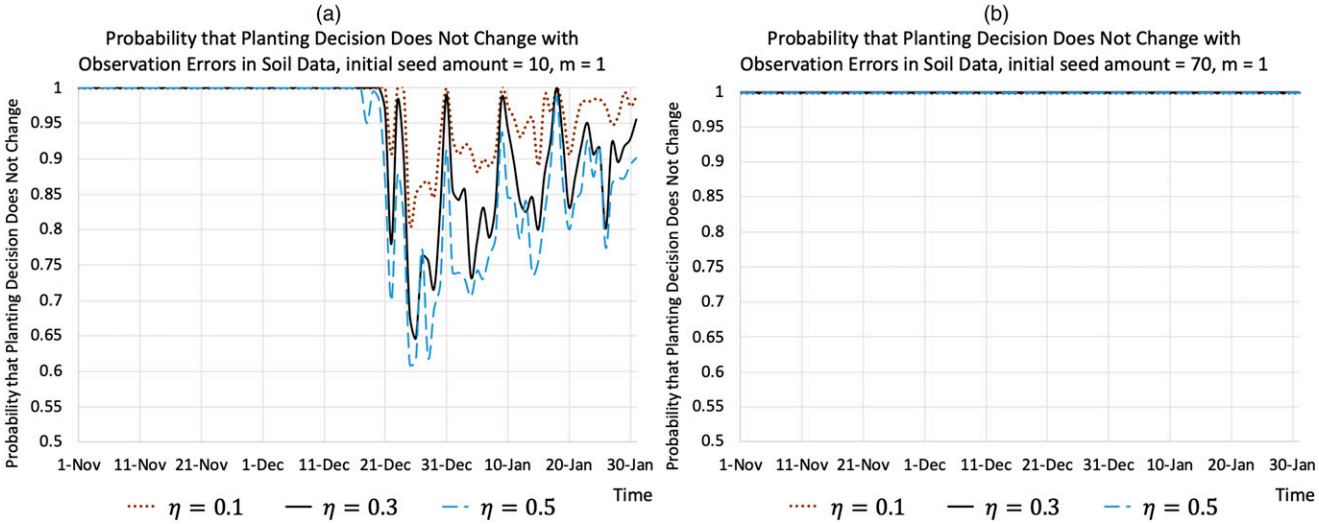
Figure 5 shows the probability that the planting decision does not change with observation errors existing in the soil water data when the initial seed amount is 10 batches (Figure 5(a)) and 70 batches (Figure 5(b)). With a smaller initial seed amount ($sd_N = 10$), the observation errors in the soil water content have a more significant impact on the planting decisions. When the initial seed amount is small ($sd_N = 10$), the probability that the planting decision does not change is decreasing in the observation error (larger deviation in planting decisions with larger η in Figure 5(a)) and the relative loss of the expected total

Table 4. Relative Loss of Expected Total Profit with Observation Errors in Soil Data ($m = 1$)

Initial seed amount	10	30	50	70
$\eta = 0.1$, %	0.0005	0.0768	0.0215	0.0000
$\eta = 0.3$, %	0.3306	0.3004	0.0745	0.0000
$\eta = 0.5$, %	0.9146	0.9458	0.2334	0.0000

Downloaded from informs.org by [137.189.217.69] on 05 October 2025, at 06:53 . For personal use only, all rights reserved.

Figure 5. (Color online) Probability that the Planting Decision Does Not Change with Observation Errors in Soil Data ($m = 1$)



profit is increasing in the observation error. When the initial seed amount is large ($sd_N = 70$), however, the observation errors in the soil water content barely influence the planting decisions or result in loss of the expected total profit.

4.6. Performance of the Secretary Heuristic

Table 5 lists the relative profit advantage of the secretary heuristic over the commonly used heuristics (negative values indicate lower average profit of the secretary heuristic). The secretary heuristic outperforms the commonly used heuristics for almost all cases except for the case of manual planting ($m = 1$), in which the performance of the secretary heuristic is slightly worse than the commonly used heuristics.

4.7. Out-of-Sample Testing

We conduct an out-of-sample test to validate our results. We use the first 10 years' data from November 2000 to April 2010 as the training set and years 2011 (November 2011 to April 2012) and 2016 (November 2016 to April 2017) as the test set. We calculate average daily rainfall data (rainfall probability and distribution of conditional precipitation), minimum and maximum daily temperature, and solar radiation of the first 10 years and each of the two years in the test set. Let $p_t^{[i],r}$ and $\Gamma_t^{[i]}(\alpha_t^{[i]}, \beta_t^{[i]}), i = 0, 1, 2; t = 1, \ldots, 181$ denote the average probability of rainfall in day t of the ith year and distribution of conditional precipitation amount in day t of the ith year ($i = 0$ represents the training set, $i = 1, 2$ represents years 2011 and 2016, and $t = 1$ represents November 1). To conduct the out-of-sample test, we search for the optimal planting thresholds using the training set. Then, for year k in the test set ($k = 1, 2$), we use the daily rainfall probability $p_t^{[k],r}, t = 1, \ldots, 181$, to generate 500 sample paths and each sample path characterizes whether it rains in each day of the year (November 1 to April 30). If it rains on day t, the precipitation amount is generated by $\Gamma_t^{[k]}(\alpha_t^{[k]}, \beta_t^{[k]})$. The expected total profit under the optimal policy is calculated by taking the average of the 500 sample-path profit values, each obtained following the plant-down-to policy with planting thresholds searched out based on the first 10 years' weather data. The expected profit under the commonly used heuristics (secretary heuristic) is calculated by taking the average of the 500 sample-path values, each obtained following the $10days - 25$mm heuristic (secretary heuristic).

Table 6 shows that, for all cases in the test set, the expected total profit under the optimal policy is higher than that under commonly used heuristics in practice ($10days - 25$mm heuristic). In most cases, the secretary heuristic outperforms the commonly used heuristics and could generate expected profit values that are close to the optimal policy, especially under mechanized planting ($m > 1$).

Table 5. Relative Expected Total Profit Advantage of the Secretary Heuristic

Planting capacity	1	2	3	4
Average soil water content, %	–0.47	4.12	3.28	2.46

Table 6. Relative Expected Total Profit Advantages of the Optimal Policy and the Secretary Heuristic in Out-of-Sample Testing

Optimal policy	2011	2016	Secretary heuristic	2011	2016
$m = 1$, %	6.67	11.26	$m = 1$, %	–1.88	4.35
$m = 2$, %	5.51	10.91	$m = 2$, %	4.98	9.61
$m = 3$, %	6.00	10.86	$m = 3$, %	4.87	9.53
$m = 4$, %	5.98	11.12	$m = 4$, %	4.42	8.93

Downloaded from informs.org by [137.189.217.69] on 05 October 2025, at 06:53 . For personal use only, all rights reserved.

4.8. Advantage of Optimal Planting Schedule Under Severe Climate Conditions

Next, we explore the impact of severe climate conditions on the advantage of the optimal planting schedule over commonly used myopic heuristics. Studies in the agriculture literature suggest that the sowing rules should be adjusted based on prior information of severe climate conditions rather than starting to plant from a constant date. One of the most commonly used sowing rules is to start planting after observing enough cumulative precipitation in a time window and keep planting afterward. The threshold of cumulative precipitation amount and the length of the time window would be adjusted based on prior information of weather data (Wolf et al. 2015). We use $[x, y]$ to denote the myopic heuristics under which the farmer would start planting at full capacity after observing x mm of precipitation in a y-day period starting from November 1 and keep planting whenever possible until the depletion of seeds or the end of the planting window, whichever occurs earlier. For each climate condition, we enumerate the myopic policies in set $M = \{[x, y] | x = 5, 10, 15, 20, 25, 30; y = 5, 15, 25, 35\}$ and pick the best one that provides the highest expected total profit. In calculating the relative advantage of the optimal policy under different climate conditions, we compare the expected total profit under the optimal policy to the best commonly used myopic heuristic in set M when the initial seed amount is between $(31 \times m)$ and $(40 \times m)$.

4.8.1. Delayed Main Rainfall. We first explore the impact of delayed main rainfall on the relative advantage of the optimal planting schedule. Our weather data shows that both the rainfall probability and precipitation tend to increase from November and starts to drop around mid-January. In recent years, however, extensive areas in Southern Africa have seen delays in the onset of crop planting of up to six weeks because of the late start of the main rainfall, and this results in a significant drop of crop production and farmers' profits (Stern and Cooper 2011). We shift the pattern of rainfall probability to indicate an early or late start of the main rainfall as shown in Figure 6(a). The *30-Day Early* (*Late*) case is generated by shifting the current rainfall pattern backward (forward) by 30 days, and thus, the peak of the rainfall probability occurs 30 days earlier (later) than the current rainfall pattern. The average rainfall probability over the rainy season ($\frac{\sum_{t=1}^{181} p_t^r}{181}$) remains the same among all cases.

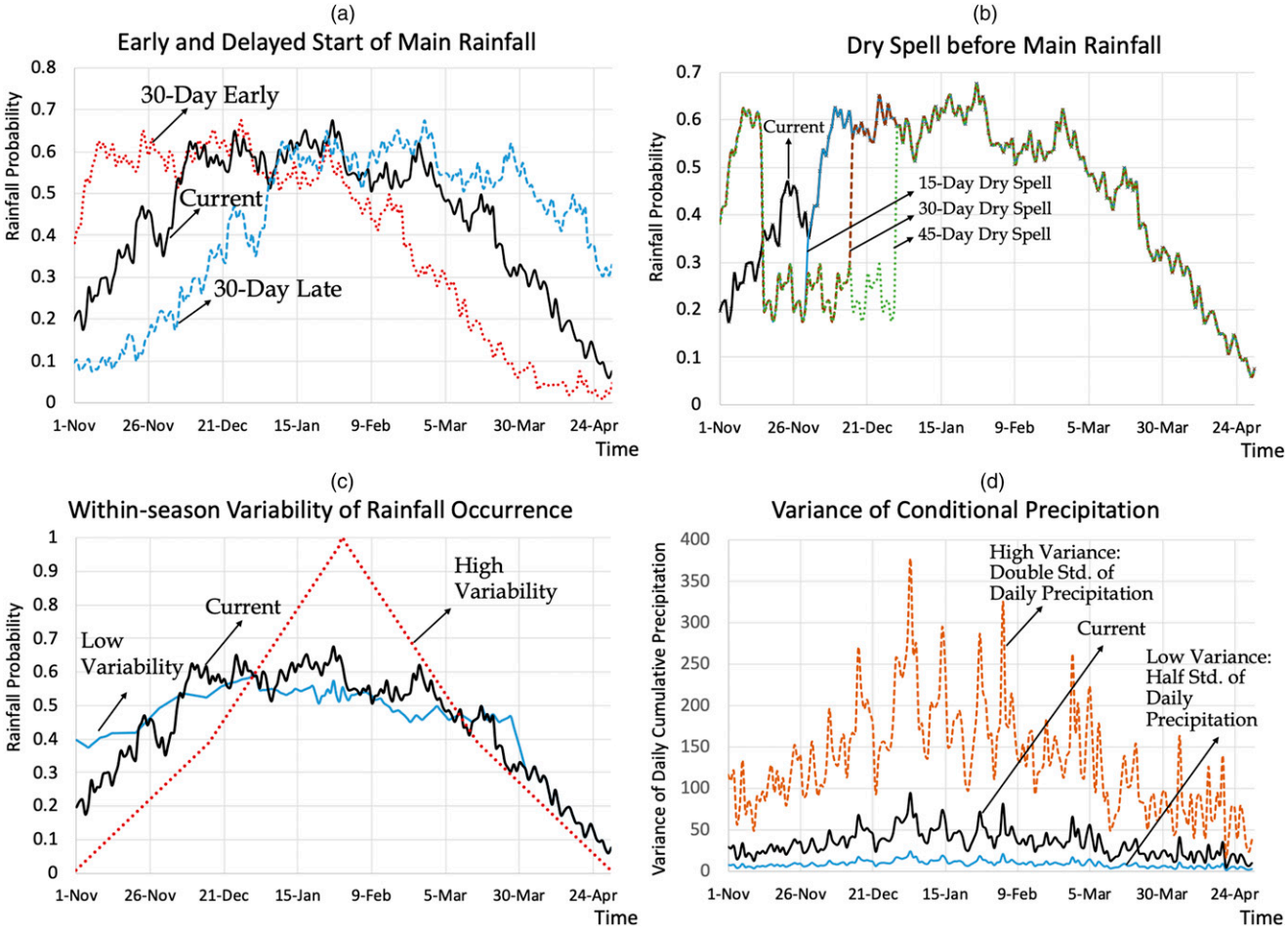
Our numerical study demonstrates that the profit loss resulting from delayed rainfall could be mitigated by applying the optimal planting schedule as presented in Table 7. The more the rainy season is delayed, the more significant advantage the optimal policy provides. With prior information about the early start of the rainy season, the farmer would start planting early under the optimal policy as well as the myopic policy so that more seeds would grow in highly moisturized soil. In our computational results, the best myopic policy with an early start of the rainy season (*30-Day Early* case) is $[5, 5]$ (start planting after observing 5 mm of cumulative precipitation in a five-day period). Under the optimal policy, the planting thresholds in early periods are low, and the farmer starts to plant early in the planting window. Therefore, the planting decisions are similar under both policies, and the relative advantage of the optimal policy is not significant. With prior information about the late start of the rainy season, under commonly used myopic heuristics, the farmer would wait for a longer time to observe cumulative precipitation, and planting starts late (best myopic policies are $[20, 35]$ under the current rainfall pattern and $[15, 35]$ under the *30-Day Late* case). Hence, all seeds may not be depleted at the end of the planting window. However, if the rainfall gets delayed, the optimal planting thresholds tend to decrease, and the farmer will start to plant at a lower soil water content. Therefore, more seeds can be planted to grow during the main rainfall, leading to a significantly higher expected profit than the commonly used heuristics.

4.8.2. Dry Spell Before Main Rainfall. Another critical agricultural issue that African countries have faced in recent years, especially with El Niño events, is that a long dry spell often occurs after early rainfall and before the main rainfall. This type of drought often aggravates circumstances. The early rainfall allows the planting process to start, but it is insufficient for crop establishment and, thus, leads to a drop in crop yields and farmers' benefit (Dennett 1987). Figure 6(b) shows three cases of early dry spells and compares them to the current rainfall pattern. The *15-Day* (*30-Day*, *45-Day*) *Dry Spell* case indicates a 15-day (30-day, 45-day) period of low rainfall occurrence after an early rainfall peak in early November.

Table 8 shows that the relative profit advantage of the optimal schedule becomes significantly larger as the expected length of the early dry spell becomes longer. With prior information about early dry spells, under commonly used myopic heuristics, the farmer would set a larger time window to observe cumulative precipitation (the best myopic policies are $[30, 25]$ under the *15-Day Dry Spell* case and $[30, 35]$ under the *45-Day Dry Spell* case) and keep planting during the dry spell. Although, in reality, a farmer might stop planting during a dry spell, this could vary from farmer to farmer. Therefore, in order to have a consistent policy, we assume that farmers continue to plant even during the dry spell in our myopic heuristics. Under the optimal schedule, however,

Downloaded from informs.org by [137.189.217.69] on 05 October 2025, at 06:53 . For personal use only, all rights reserved.

Figure 6. (Color online) Rainfall Patterns Under Severe Climate Conditions



planting may not continue after early rainfall if a dry spell is encountered in the following periods and results in low soil water content, and thereby low expected profits from planting seeds in the periods after the early rainfall. When the early dry spell becomes longer, the optimal planting thresholds of the early periods become higher, and planting occurs at a higher soil water content and a later period. When the initial soil water content is low, the farmer using the optimal policy starts planting later when the dry spell is about to end. If the initial soil water content is high and the farmer starts planting when the early rainfall is observed, the farmer may stop planting during the dry spell if low soil water content is expected and continue to plant later when the main rainfall occurs. Therefore, the profit loss resulting from the longer early dry spell could be mitigated by adopting the optimal schedule.

4.8.3. Within-Season Variability of Rainfall. The within-season variability of rainfall usually leads to lower crop yields and, thus, lower profits of farmers (Stern and Cooper 2011). In Africa, the distribution of daily precipitation is often highly skewed with 23% of rainy days contributing 80% of total rainfall (Dennett 1987). In our model, we consider the within-season variability of rainfall occurrence and the conditional precipitation amount. We use the flatness of the rainfall probability pattern to indicate within-season variability of the rainfall occurrence as in Figure 6(c). The average rainfall probability over the rainy season ($\frac{\sum_{t=1}^{181} p_t^r}{181}$) remains the same among all rainfall patterns. In Figure 6(d), we double (halve) the standard deviation of the daily conditional precipitation amount to represent the high (low) variability of the precipitation amount and keep the mean of daily conditional precipitation (α_t/β_t for each period t) the same through all scenarios.

Table 9 shows that the relative profit advantage of the optimal planting schedule increases as the within-season variability of rainfall occurrence increases. Under such conditions, the optimal planting thresholds of the early

Table 7. Relative Expected Total Profit Advantage of the Optimal Policy with Early or Delayed Rainfall

Planting capacity	1	2	3	4
30-day early, %	1.12	1.21	1.16	1.18
Current rainfall, %	4.68	4.69	5.08	4.97
30-day late, %	13.72	12.49	13.75	13.37

Downloaded from informs.org by [137.189.217.69] on 05 October 2025, at 06:53 . For personal use only, all rights reserved.

Table 8. Relative Expected Total Profit Advantage of the Optimal Policy with Dry Spell Before Main Rainfall

Planting capacity	1	2	3	4
15-day dry spell, %	6.71	6.87	7.48	7.04
30-day dry spell, %	10.94	11.74	12.25	12.19
45-day dry spell, %	20.96	21.72	20.90	22.13

periods decrease and those of later periods increase. The farmer starts to plant earlier so that more seeds can be planted before the heavy rain comes. Therefore, more seeds can grow with high soil water content, and this leads to a higher total profit. Under commonly used heuristics, a higher variability of rainfall occurrence leads to a larger time window to observe cumulative precipitation (best myopic policies are [30, 25] with low variability and [5, 35] with high variability), and planting starts later as the probabilities of rainfall occurrence in early periods are lower. When the rainfall peaks, fewer seeds have been planted, and the planting process may be impeded as farmers cannot plant when it rains. Therefore, with increasing variability of rainfall occurrence, the expected total profit under commonly used heuristics tends to become lower.

Table 10 shows that the relative profit advantage of the optimal policy increases as the variability of daily conditional precipitation increases. Higher variability of precipitation harms the profit as seeds grow with more unstable soil water content. Under commonly used heuristics, the farmer requires less cumulative precipitation to start planting with higher variability of conditional precipitation (best myopic policies are [25, 35] with low variability of precipitation and [20, 35] with normal and high variability of precipitation). Under the optimal policy, the optimal planting thresholds decrease with higher variability of precipitation, and planting starts at a lower soil water content. Furthermore, planting is interrupted at very low soil water content after consecutive days of low precipitation. This reserves seeds for better planting conditions and, thus, mitigates yield loss even more.

5. Model Variants

In this section, we study two variants of the crop planting model: (1) planting with fixed cost and (2) planting with rainfall occurrence modeled as a DTMC.

5.1. Planting with Fixed Cost

Farmers may incur a fixed cost on external labor or machines for planting in addition to variable costs. We discuss the planting schedule in this case where the farmer incurs a fixed cost denoted by K when she decides to plant in period t. Let $\delta(.)$ denote the indicator function, where $\delta(0) = 0$ and $\delta(x) = 1$ if $x > 0$. If i_t batches of seeds are planted in period t, the farmer would incur a fixed cost of $K\delta(i_t)$, and the expected profit from planting in period t would be $i_t G_t(sw_t) - K\delta(i_t)$. We use the superscript K to represent the planting schedule problem with fixed cost of K at planting.

$$\begin{aligned} V_t^{K,m}(sd_t, sw_t) &= p_t^r \mathbb{E}_{prec_t} V_{t-1}^{K,m}\big(sd_t, \omega_t(sw_t, prec_t, \boldsymbol{\lambda}_t)\big) \\ &\quad + \big(1 - p_t^r\big) \max_{i_t=0,\ldots,\min\{sd_t,m\}} \Big\{ V_{t-1}^{K,m}(sd_t - i_t, \\ &\quad \omega_t(sw_t, 0, \boldsymbol{\lambda}_t)) + i_t G_t(sw_t) - K\delta(i_t) \Big\}, t = 1, \ldots, N \\ V_t^{K,m}(sd_0, sw_0) &= r_v \times sd_0. \end{aligned} \tag{3}$$

The optimality of the threshold-type policy no longer holds with a fixed cost of planting and limited planting capacity except for the manual planting case with $m = 1$. This is consistent with results in inventory literature (Gallego and Scheller-Wolf 2000). Following a similar procedure as in Gallego and Scheller-Wolf (2000), we can partially characterize the optimal policy under planting capacity m ($m > 1$) using time-dependent bands $(\Theta_t^{m,L}(sw_t), \Theta_t^{m,H}(sw_t))$: for system state $s_t^m = (sd_t, sw_t)$ in sunny period t, there exist two thresholds, $\Theta_t^{m,L}(sw_t), \Theta_t^{m,H}(sw_t)$, such that the farmer should plant at full capacity if $sd_t > \Theta_t^{m,H}(sw_t)$ and not plant if $sd_t \le \Theta_t^{m,L}(sw_t)$.

5.2. Rainfall Occurrence in DTMC

In this section, the occurrence of rainfall is modeled as a second-order DTMC as in the literature (Srikanthan and McMahon 2001). Then, the first-order chain can be regarded as a special case of the second-order chain.

Let $H_t = (X_{t+1}, X_{t+2}) \in \{(X_{t+1}, X_{t+2}) | X_{t+1}, X_{t+2} = R, S\}$ denote the weather history observed at the beginning of period t, where X_t is the weather in period t: rainy (R) or sunny (S). Given the weather history, we use $p_t^r(H_t) = Prob\{X_t = R | H_t\}$ to denote the probability that

Table 9. Relative Expected Total Profit Advantage of the Optimal Policy with Varying Variability of Rainfall Occurrence

Planting capacity	1	2	3	4
Low variability, %	4.26	3.84	4.27	4.52
Current rainfall, %	4.68	4.69	5.08	4.97
High variability, %	32.44	34.27	32.16	32.35

Table 10. Relative Expected Total Profit Advantage of the Optimal Policy with Varying Variability of Conditional Precipitation

Planting capacity	1	2	3	4
Half standard of precipitation, %	3.63	2.87	3.37	4.12
Normal precipitation variability, %	4.68	4.69	5.08	4.97
Double standard of precipitation, %	6.52	5.54	6.32	5.52

Downloaded from informs.org by [137.189.217.69] on 05 October 2025, at 06:53 . For personal use only, all rights reserved.

period t is rainy and with probability $(1 - p_t^r(H_t))$ period t is sunny. We use superscript MC to represent the planting schedule problem with the rainfall occurrence in a DTMC and we can show that the optimality of the plant-down-to policy is preserved.

$$\begin{aligned}
&V_t^{MC,m}(sd_t, sw_t, H_t) \\
&\quad = p_t^r(H_t)\mathbb{E}_{prec_t} V_{t-1}^{MC,m}\big(sd_t, \omega_t\big(sw_t, prec_t, \boldsymbol{\lambda}_t\big), H_{t-1} \\
&\quad = (R, X_{t+1})) + \big(1 - p_t^r(H_t)\big) \\
&\qquad \times \max_{i_t=0,\ldots,\min\{sd_t,m\}} \Big\{ V_{t-1}^{MC,m}(sd_t - i_t, \omega_t(sw_t, 0, \boldsymbol{\lambda}_t), \\
&\qquad H_{t-1} = (S, X_{t+1})) + i_t G_t(sw_t) \Big\}, t = 1, \ldots, N \\
&V_0^{MC,m}(sd_0, sw_0, H_0) = r_v \times sd_0. \qquad (4)
\end{aligned}$$

6. Concluding Remarks

The increase in human population has brought a lot of attention to agriculture in African countries. With the average planting yields far below the developing world average, farmers in Africa need to adopt advanced planting techniques to increase crop yields and their profits. Further, agriculture in Africa faces severe issues because of delay in the main rainfall, long dry spells before the main rainfall, and high within-season variability of rainfall. As climate conditions become more severe, farmers' crop profits are seriously harmed.

In this paper, we study the planting schedule problem of a single crop under rainfall uncertainty as a finite-horizon stochastic dynamic program. Planting early could ensure that more seeds can be planted throughout the planting horizon, but low soil water content in early periods could result in low profit of planting. We show that the optimal planting schedule is a time-dependent, threshold-type policy, in which the farmer should plant down to the optimal threshold of seed amount.

In practice, farmers start to plant each year after observing enough cumulative rainfall. Utilizing field weather data collected from nine countries in Southern Africa and the well-known crop yield simulator DSSAT, we show that adopting the optimal schedule could significantly improve the farmer's expected total profit. We also develop a heuristic based on the traditional secretary problem and show that it outperforms the commonly used heuristics in most cases. Furthermore, our results demonstrate that the risk from severe climate conditions can be significantly mitigated by adopting the optimal planting schedule. The more severe the climate conditions, the higher the relative profit advantage of the optimal planting schedule.

One limitation to the implementation of the optimal policy is the ready availability of soil moisture and weather data in all locations. Government and entrepreneurs in Africa are making efforts to provide such information through cellphones (Klein 2015). Further, low-cost soil moisture meters are being developed for widespread usage. These changes would facilitate immediate application of optimal seeding policies that utilize soil moisture in the near future.

6.1. Limitations and Extensions

Finally, we discuss the limitations of our model and provide possible extensions for future research.

1. Our model captures the capacity constraint through the planting capacity. More detailed models that capture the capacity constraint in a more refined manner and in other aspects, such as weeding, irrigation, and harvesting, could be useful.

2. We discuss the planting schedule of a single crop. It is important for farmers to determine what they should sow when there are multiple crops with different characteristics of productivity. In addition, crop rotation has an important impact on the yields of seeds, and in such cases, it needs to be incorporated.

3. We assume fixed planting capacity and fixed initial seed amount in our model and do not consider any budget constraints that a farmer might face for seed procurement, automation, fertilizers, or pest control. A possible extension could be to explore the optimal allocation of the financial budget between seeds and mechanization.

4. We mostly consider rainwater to irrigate the land in our model. It would be impactful to discuss the optimal irrigation policy when irrigation incurs cost.

5. We model the planting schedule problem in a dynamic programming framework. Other models, such as robust dynamic programming or a Bayesian approach, could also be applied to characterize the optimal planting policy.

6. In this paper, we only consider planting schedule optimization as a means to improve farm productivity. However, subsidies and minimum support prices can also have an important impact. A future research area could be around integrating these topics.

Acknowledgments

The authors gratefully acknowledge the help of the editor-in-chief, the associate editor, and the anonymous referees for their constructive comments. The authors thank the AGCO Africa team for help with model validation. They also thank all seminar participants at Chicago Booth, University of Southern California Marshall, Cornell Johnson and Cornell Operations Research and Information Engineering, Hong Kong Polytechnic University, Purdue University, Clemson University, Singapore Management University, University of New Hampshire, and Saint Louis University for their feedback.

Downloaded from informs.org by [137.189.217.69] on 05 October 2025, at 06:53 . For personal use only, all rights reserved.

References

Alizamir S, Iravani F, Mamani H (2019) An analysis of price vs. revenue protection: Government subsidies in the agriculture industry. *Management Sci.* 65(1):32–49.

Allen SJ, Schuster EW (2004) Controlling the risk for an agricultural harvest. *Manufacturing Service Oper. Management* 6(3):225–236.

Allen CJB, Catchot D, Cook D, Crow W, Dean J, Gore J, Layton B, Little N, MacGown J, Musser F (2017) 2018 insect control guide for agronomic crops. Technical report, Mississippi State University.

Ata B, Lee D, Sönmez E (2019) Dynamic volunteer staffing in multicrop gleaning operations. *Oper. Res.* 67(2):295–314.

Boyabatlı O, Nasiry J, Zhou Y (2019) Crop planning in sustainable agriculture: Dynamic farmland allocation in the presence of crop rotation benefits. *Management Sci.* 65(5):2060–2076.

Clark C, Long M, Anderson CL (2009) Labor constraints in the sub-Saharan agricultural market. Technical report, Evans School Policy Analysis and Research, Seattle, WA.

Dawande M, Gavirneni S, Mehrotra M, Mookerjee V (2013) Efficient distribution of water between head-reach and tail-end farms in developing countries. *Manufacturing Service Oper. Management* 15(2):221–238.

Dennett MD (1987) Variation of rainfall—The background to soil and water management in dryland regions. *Soil Use Management* 3(2):47–51.

Federgruen A, Zipkin P (1986) An inventory model with limited production capacity and uncertain demands II. The discounted-cost criterion. *Math. Oper. Res.* 11(2):208–215.

Food and Agriculture Organization (2009) The special challenge for sub-Saharan Africa. Accessed August 27, 2020, http://www.fao.org/fileadmin/templates/wsfs/docs/Issues_papers/HLEF2050_Africa.pdf.

Freeman P (1983) The secretary problem and its extensions: A review. *Internat. Statist. Rev.* 51(2):189–206.

Gallego G, Scheller-Wolf A (2000) Capacitated inventory problems with fixed order costs: Some optimal policy structure. *Eur. J. Oper. Res.* 126(3):603–613.

Grant BL (2018) When to fertilize plants: Best times for application of fertilizer. Accessed August 27, 2020, https://www.gardeningknowhow.com/garden-how-to/soil-fertilizers/best-times-to-apply-fertilizer.htm.

Graves SC (1981) A review of production scheduling. *Oper. Res.* 29(4):646–675.

Hu M, Liu Y, Wang W (2019) Socially beneficial rationality: The value of strategic farmers, social entrepreneurs, and for-profit firms in crop planting decisions. *Management Sci.* 65(8):3654–3672.

Jones JW, Hoogenboom G, Porter CH, Boote KJ, Batchelor WD, Hunt L, Wilkens PW, Singh U, Gijsman AJ, Ritchie JT (2003) The DSSAT cropping system model. *Eur. J. Agronomy* 18(3–4):235–265.

Kazaz B (2004) Production planning under yield and demand uncertainty with yield-dependent cost and price. *Manufacturing Service Oper. Management* 6(3):209–224.

Kazaz B, Webster S (2011) The impact of yield-dependent trading costs on pricing and production planning under supply uncertainty. *Manufacturing Service Oper. Management* 13(3):404–417.

Kings S (2019) Maize vs time: The race to feed SA. Accessed August 27, 2020, https://mg.co.za/article/2019-01-25-00-maize-vs-time-the-race-to-feed-sa/.

Klein K (2015) Cell phones monitor water, soil on African farms. Accessed August 27, 2020, https://blogs.scientificamerican.com/guest-blog/cell-phones-monitor-water-soil-on-african-farms-q-amp-a/.

Lopes C (2014) We need more agribusiness in Africa. Accessed August 27, 2020, http://www.un.org/africarenewal/magazine/special-edition-agriculture-2014/we-need-more-agribusiness-africa.

Lowe TJ, Preckel PV (2004) Decision technologies for agribusiness problems: A brief review of selected literature and a call for research. *Manufacturing Service Oper. Management* 6(3):201–208.

Maatman A, Schweigman C, Ruijs A, van Der Vlerk MH (2002) Modeling farmers' response to uncertain rainfall in Burkina Faso: A stochastic programming approach. *Oper. Res.* 50(3):399–414.

McGuire S (2016) Africa's informal seed system needs to be brought in from the cold. Accessed August 27, 2020, https://theconversation.com/africas-informal-seed-system-needs-to-be-brought-in-from-the-cold-54609.

National Agricultural Statistics Service (2010) Usual planting and harvesting dates for U.S. field crops. Technical report, United States Department of Agriculture, Washington, DC.

Obasanjo O (2012) How Africa could feed the world. Accessed August 27, 2020, http://globalpublicsquare.blogs.cnn.com/2012/11/06/how-africa-could-feed-the-world/.

SOLA (2013) South Africa's solar resource compared with the rest of the world. Accessed August 27, 2020, https://www.solafuture.co.za/news/south-africas-solar-resource-compared-to-the-rest-of-the-world/.

Srikanthan R, McMahon T (2001) Stochastic generation of annual, monthly and daily climate data: A review. *Hydrology Earth Systems Sci.* 5(4):653–670.

Srivastava AK, Mboh CM, Gaiser T, Webber H, Ewert F (2016) Effect of sowing date distributions on simulation of maize yields at regional scale—A case study in central Ghana, West Africa. *Agricultural Systems* 147:10–23.

Stern RD, Cooper PJM (2011) Assessing climate risk and climate change using rainfall data—A case study from Zambia. *Experiment. Agriculture* 47(2):241–266.

Swaminathan JM (2018) AGCO: Innovating the agricultural value chain in Africa. Case study, Kenan-Flagler Business School, University of North Carolina at Chapel Hill, Chapel Hill.

Swaminathan J, Zhang Y (2020) Improving crop yields through better planting schedules. Swaminathan J, Deshpande V, eds. *Responsible Business Operations* (Springer Nature Switzerland AG, Cham, Switzerland).

Tadross M, Suarez P, Lotsch A, Hachigonta S, Mdoka M, Unganai L, Lucio F, Kamdonyo D, Muchinda M (2009) Growing-season rainfall and scenarios of future change in southeast Africa: Implications for cultivating maize. *Climate Res.* 40(2–3):147–161.

Tan B, Çömden N (2012) Agricultural planning of annual plants under demand, maturation, harvest, and yield risk. *Eur. J. Oper. Res.* 220(2):539–549.

Tang CS, Wang Y, Zhao M (2015) The implications of utilizing market information and adopting agricultural advice for farmers in developing economies. *Production Oper. Management* 24(8):1197–1215.

Tang CS, Wang Y, Zhao M (2018) The impact of input-vs. output-based farm subsidies on farmer welfare and income inequality in developing economies. Working paper, UCLA Anderson School of Management, Los Angeles.

United Nations (2009) Food production must double by 2050 to meet demand from world's growing population, innovative strategies needed to combat hunger, experts tell second committee. Accessed August 27, 2020, http://www.un.org/press/en/2009/gaef3242.doc.htm.

Wolf J, Ouattara K, Supit I (2015) Sowing rules for estimating rainfed yield potential of sorghum and maize in Burkina Faso. *Agricultural Forest Meteorology* 214:208–218.

Zinyengere N, Crespo O, Hachigonta S, Tadross M (2015) Crop model usefulness in drylands of southern Africa: An application of DSSAT. *South African J. Plant Soil* 32(2):95–104.